Experiment-3

Name of the experiment: Determination of tensile strength of rock samples by Brazilian method

INTRODUCTION

Tensile strength may be defined as the maximum stresses developed in a specimen in a tension test performed to rupture it. In rock mechanics knowledge of tensile strength of rock is important in designing roofs and domes of underground openings. A rock slab or beam subjected to bending also experiences a tensile stress. Although rock is weak in tension, an average tensile strength of rock has been found to be one tenth of its compressive strength. Although the principle of measurement of tensile strength is to apply an axial tensile force to a rock specimen the method is not successful due to difficulties in making specimen of rock samples for the test. With rock, it is very difficult to make specimen of such a shape. An alternative is to grip the cylindrical sample at two ends with some fixing material, and then to apply the tensile force at the two ends. Due to the gripping difficulties, the tensile strength of rock is estimated by indirect method (Brazilian test).

SCOPE

This test is intended to measure the uniaxial tensile strength of prepared rock specimens indirectly by the Brazil test. The justification for the test is based on the experimental fact that most rocks in biaxial stress fields fail in tension at their uniaxial tensile strength when one principal stress is tensile and the other finite principal stress is compressive with a magnitude not exceeding three times that of the tensile principal stress.

APPARATUS

(a) Two steel loading jaws designed so as to contact a disc-shaped rock sample at diametrically-opposed surfaces over an arc of contact of approx. 10° at failure. The suggested apparatus to achieve this is illustrated in Figure 1. The critical dimensions of the apparatus are the radius of curvature of the jaws, the clearance and length of the guide pins coupling the two curved jaws and the width of the jaws. These are as follows: Radius of jaws - l.5 × specimen radius; guide pin clearance - permit rotation of one jaw relative to the other by 4×10^{-3} rad out of plane of the apparatus (25 mm penetration of guide pin with 0.1 mm clearance); width of jaws - l.1 × specimen thickness. The remaining dimensions can be scaled from Figure 1. The upper jaw contains a spherical seating conveniently formed by a 25 mm diameter half-ball bearing.

(b) Double thickness (0.2 - 0.4mm) adhesive paper strip (masking tape) with a width equal to or slightly greater than the specimen thickness.

(c) A suitable machine for applying and measuring compressive loads to the specimen. It shall be of sufficient capacity and be capable of applying load at a rate conforming to the requirements set out in section.

(d) Equipments for measuring dimensions of the rock sample

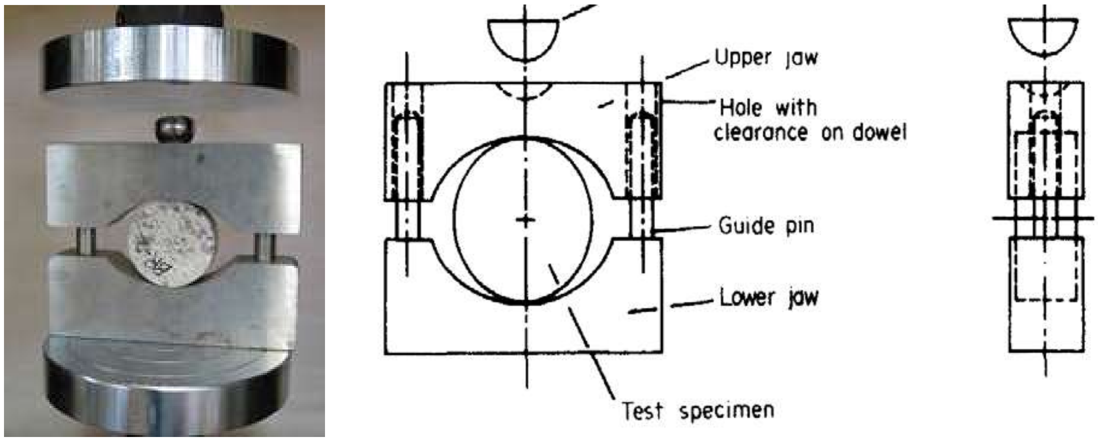

Fig. 1: Apparatus for Brazilian test

PROCEDURE

(a) The test specimens should be cut and prepared using clean water. The cylindrical surfaces should be free from obvious tool marks and any irregularities across the thickness of the specimen should not exceed 0.025 mm. End faces shall be fiat to within 0.25 mm and square and parallel to within 0.25°.

(b) The specimen diameter shall not be less than NX core size, approximately 54mm, and the thickness should be approximately equal to the specimen radius.

(c) The test specimen shall be wrapped around" its periphery with one layer of the masking tape and mounted squarely in the test apparatus such that the curved platens load the specimen diametrally with the axes of rotation for specimen and apparatus coincident.

(d) Load on the specimen shall be applied continuously at a constant rate such that failure in the weakest rocks occurs within 15-30 s. A loading rate of 200 N/s is recommended (Figure 2).

(e) The number of specimens per sample tested should be determined from practical considerations, but normally ten are recommended.

CALCULATIONS

The tensile strength of the specimen σ_t, shall be calculated by the following formula:

$$\sigma_t = 0.636\, P/Dt \text{ (MPa)}$$

where P is the load at failure(N), D is the diameter of the test specimen (in mm), t is the thickness of the test specimen measured at the center (in mm).

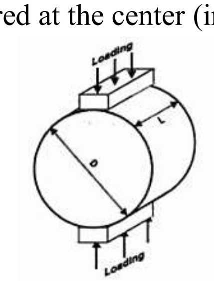

Fig. 2: Brazilian testing arrangement for determining indirect tensile strength

Tensile strength test is to verify tensile strength of rock samples or its resistance against fracturing. Direct tensile test on rock sample is relatively difficult to undertake and Brazilian test offers an indirect method to measure tensile strength. The term ‘indirect tensile’ implies that load is applied under compression. Figure 3 shows typical mode of failure associated with tensile stress.

Figure 3: Mode of failure of rock samples in Brazilian test

Observations:

Sample No.	Rock Type	Diameter	Avg. Diameter	Thickness	Avg. Thickness	Weight	Breaking load	Total time taken	Circumferential area	Stress rate or Rate of	Density	Avg. Density	Type of failure	Tensile strength	Avg. Tensile strength

References:

International Society for Rock Mechanics Suggested Methods "Rock Characterization, Testing and Monitoring" Editor E.T. Brown, Pergamon Press 1981.